Electronic Devices & Circuits Laboratory (EE2301)

Experiment 8: NMOS Linear Voltage Regulator using a 741 Op-Amp:

1.2 V Regulated Output from a 3 to 5 V Supply

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1 Objective

The specific experimental objectives of this laboratory exercise are:

1. To build a linear voltage regulator using an LM741 op-amp driving the gate of an N-channel power MOSFET connected as a source follower.
2. To hold the output steady at $V_{OUT} = 1.200\,\text{V}$ while the input supply varies from 3.0 V to 5.0 V.
3. To measure the **line regulation** in mV/V and observe the feedback-driven gate bias tracking compensating for pass transistor channel-length modulation.
4. To characterize the **load regulation** across load currents up to $\approx 118\,\text{mA}$ and extract the closed-loop output resistance R_{OUT} (mV/A or mΩ).
5. To measure the **dropout voltage** at full load ($R_L \approx 10.2\,\Omega$) and relate it directly to the MOSFET's linear triode on-resistance $R_{DS(\text{on})}$ and op-amp swing limits.
6. To measure the power dissipation P_{diss} and efficiency (η_{pass} and η_{total}), and understand why linear regulator efficiency is limited to V_{OUT}/V_{IN}.
7. To check the feedback stability on an oscilloscope with different output capacitors ($10\,\mu\text{F}, 47\,\mu\text{F}, 94\,\mu\text{F}$) and check for any oscillation.
8. To answer all eleven analytical discussion questions specified in Section 9 of the laboratory manual.

2 Theoretical Background and Operating Principles

2.1 Closed-Loop Negative Feedback Action

The regulator uses negative feedback to keep the output voltage V_{OUT} equal to a fixed reference voltage $V_{REF} = 1.200\,\text{V}$. The LM741 op-amp acts as the error amplifier:

- The non-inverting input (Pin 3) is connected to the reference voltage $V_{REF} = 1.200\,\text{V}$.
- The inverting input (Pin 2) directly senses the regulated output voltage V_{OUT} at the source of M_1.
- The op-amp output (Pin 6) drives the gate of the N-channel power MOSFET.

Because the MOSFET operates as a source follower (common drain), the output voltage follows the gate voltage shifted downward by the gate-to-source voltage:

$$V_{OUT} = V_G - V_{GS} \tag{1}$$

If the output voltage V_{OUT} drops slightly (for example when load current increases or V_{IN} decreases), the differential input voltage $V_{id} = V_{REF} - V_{OUT}$ becomes positive. Because the 741 op-amp has a

very high open-loop gain ($A \approx 2 \times 10^5$), its output quickly drives the gate voltage V_G higher. This increases $V_{GS} = V_G - V_{OUT}$, so the MOSFET provides more drain current to the load, bringing V_{OUT} back up to V_{REF}. On the other hand, if V_{OUT} rises above V_{REF}, the op-amp lowers V_G, which reduces V_{GS} and lowers the current.

With unity feedback ($\beta = 1$), the closed-loop output voltage and error are given by:

$$V_{OUT} = V_{REF}\left(\frac{T}{1+T}\right), \quad \text{Error} = \frac{V_{REF}}{1+T} \approx \frac{V_{REF}}{A} \approx \frac{1.200\,\text{V}}{2 \times 10^5} \approx 6\,\mu\text{V} \tag{2}$$

Because the loop gain $T \approx A \approx 2 \times 10^5$ is very large, the theoretical steady-state error is only about $6\,\mu$V.

2.2 Small-Signal Line Sensitivity and Closed-Loop Output Resistance

Applying small-signal analysis to the source follower within the negative feedback loop:

- **Line Sensitivity (dV_{OUT}/dV_{IN}):** Writing KCL at the output node with small-signal gate voltage $v_g = -Av_{out}$:

$$i_d = g_m v_{gs} + \frac{v_{ds}}{r_o} = g_m(-Av_{out} - v_{out}) + \frac{v_{in} - v_{out}}{r_o} = \frac{v_{out}}{R_L} \tag{3}$$

  Solving for the line sensitivity yields:

$$\frac{dV_{OUT}}{dV_{IN}} = \frac{1}{r_o\left[g_m(1+A) + \frac{1}{r_o} + \frac{1}{R_L}\right]} \approx \frac{1}{r_o g_m(1+A)} \tag{4}$$

  For our setup ($A \approx 2 \times 10^5$, $g_m \approx 0.85\,\text{A/V}$, $r_o \approx 50\,\Omega$):

$$\frac{dV_{OUT}}{dV_{IN}} \approx \frac{1}{50 \times 0.85 \times 2 \times 10^5} \approx 1.18 \times 10^{-7}\,\text{V/V} = 0.00012\,\text{mV/V} \tag{5}$$

  In the laboratory, this theoretical microvolt change cannot be observed directly because it is hidden by reference drift, op-amp offset drift, and the multimeter resolution limit (0.1 mV).

- **Closed-Loop Output Resistance (R_{OUT}):** Applying an external test current source i_{test} with $v_{in} = 0$ and $v_{ref} = 0$:

$$i_{test} = -g_m v_{gs} + \frac{v_{test}}{r_o} = g_m(1+A)v_{test} + \frac{v_{test}}{r_o} \tag{6}$$

  yielding the closed-loop output resistance:

$$R_{OUT} = \frac{v_{test}}{i_{test}} = \frac{1}{g_m(1+A) + \frac{1}{r_o}} \approx \frac{1}{g_m A} \tag{7}$$

  Evaluating numerically:

$$R_{OUT} \approx \frac{1}{0.85\,\text{A/V} \times 2 \times 10^5} \approx 5.88\,\mu\Omega = 0.00588\,\text{mV/A} \tag{8}$$

  On the breadboard, our measured output resistance is higher (around $9 - 10\,\text{m}\Omega$) because of contact and jumper wire resistances.

2.3 Why the LM741 Requires Dedicated Auxiliary Supplies

The 741 op-amp cannot be powered directly from the $3 - 5\,\text{V}$ input rail because of two practical limitations:

1. **Gate-Drive Voltage Headroom:** For an NMOS source follower delivering $V_{OUT} = 1.200\,\text{V}$, the gate must sit at:

$$V_G = V_{OUT} + V_{GS} \approx 1.2\,\text{V} + (0.9\,\text{V to } 1.0\,\text{V}) \approx 2.1\,\text{V to } 2.2\,\text{V} \tag{9}$$

   The 741 output stage cannot swing closer than about 2 V to its positive rail ($V_{OH} \leq V^+ - 2\,\text{V}$). Powered from $V_{IN} = 3.0\,\text{V}$, the maximum gate voltage it could provide is only 1.0 V. But to keep $V_{OUT} = 1.2\,\text{V}$, the gate must sit at around $2.1 - 2.2\,\text{V}$. Therefore, powered from 3 V, it cannot turn the MOSFET on.

2. **Input Common-Mode Range (CMR):** The 741 input common-mode range also stops about 2 V above its negative rail. If the negative rail is grounded (0 V), the minimum input voltage it can sense is 2.0 V. It cannot sense a 1.200 V reference or feedback voltage.

So we power the op-amp with separate $\pm 12\,\text{V}$ (or $\pm 10\,\text{V}$) supplies to give enough headroom for both inputs and output.

2.4 Role of the Minimum-Load Bleeder Resistor

An NMOS source follower can supply current to the load, but it **cannot sink current**. If the load resistor is disconnected and there is no bleeder resistor:

- The output node has no path to discharge to ground.
- Small leakage currents or capacitive coupling from the gate can charge up C_{OUT}.
- Even though the op-amp pulls V_G low to turn the MOSFET off, it cannot pull charge out of the source terminal.
- As a result, V_{OUT} can sit higher than V_{REF}.

Connecting a permanent bleeder resistor ($1.0 - 1.2\,\text{k}\Omega$) draws a continuous baseline current:

$$I_{bleeder} = \frac{V_{OUT}}{R_{bleeder}} = \frac{1.200\,\text{V}}{1.2\,\text{k}\Omega} = 1.00\,\text{mA} \tag{10}$$

which provides a discharge path and ensures V_{OUT} stays regulated at 1.200 V even when no external load is connected.

2.5 Dropout Voltage Physics

When V_{IN} drops close to 1.2 V, the drain-to-source voltage $V_{DS} = V_{IN} - V_{OUT}$ becomes very small. To keep supplying the required load current, the op-amp output saturates near its positive rail ($V_G \approx 4.39\,\text{V}$). With the gate driven high, the MOSFET enters the ohmic (triode) region and acts like a small resistor $R_{DS(\text{on})}$. The output then follows V_{IN} through a voltage divider:

$$V_{OUT} = V_{IN}\left(\frac{R_L}{R_L + R_{DS(\text{on})}}\right) \tag{11}$$

The dropout voltage is defined per the manual as the V_{IN} at which V_{OUT} drops by 1% (12 mV) below its regulated value ($V_{OUT} \leq 1.1882\,\text{V}$). The required headroom is:

$$V_{dropout,\text{headroom}} = V_{IN,\text{dropout}} - V_{OUT} \approx I_{LOAD} \cdot R_{DS(\text{on})} + V_{DS,\text{sat}} \tag{12}$$

2.6 Efficiency Limitations: Linear vs. Switching Regulators

Because the load current flows directly through the series pass transistor ($I_{IN} \approx I_{LOAD}$), the linear regulator efficiency is fundamentally bounded by:

$$\eta_{pass} = \frac{P_{OUT}}{P_{IN}} = \frac{V_{OUT} \cdot I_{LOAD}}{V_{IN} \cdot I_{IN}} \approx \frac{V_{OUT}}{V_{IN}} \tag{13}$$

The difference voltage ($V_{IN} - V_{OUT}$) appears across the MOSFET channel, and the power is lost as heat:

$$P_{diss} = (V_{IN} - V_{OUT}) \cdot I_{LOAD} \tag{14}$$

For $V_{OUT} = 1.2\,\text{V}$, the maximum theoretical efficiency is 40.0% at $V_{IN} = 3.0\,\text{V}$ and drops to 24.0% at $V_{IN} = 5.0\,\text{V}$. In comparison, a switching buck converter turns its transistor fully ON (where $V_{DS} \approx 0$) or fully OFF (where $I = 0$), and uses an inductor and capacitor filter to average the voltage ($V_{OUT} = D \cdot V_{IN}$), reaching typical efficiencies of $80 - 95\%$.

3 Components and Laboratory Hardware Setup

The hardware components and test instruments specified in the manual and utilized during the experiment are listed in Table 1.

Table 1: Hardware Components and Test Instruments Utilized (Lab Manual Section 3).

Item	**Specification / Part Number**	**Qty**	**Role / Function**
Error Amplifier	LM741CN DIP-8 Operational Amplifier	1	Powered from $\pm 12\,\text{V}$ (or $\pm 10\,\text{V}$) aux supplies
Pass Transistor	Vishay Si9926BDY Dual N-MOSFET (SO-8 on DIP)	1	$V_{DS} \leq 20\,\text{V}$, $V_{th} \approx 0.7 - 0.9\,\text{V}$, low $R_{DS(\text{on})}$
Shunt Reference	TL431 Precision Shunt Regulator (diode mode)	1	Stable 2.5 V zener reference
Potentiometer	$10\,\text{k}\Omega$ multi-turn cermet trimmer potentiometer	1	Calibrated to $V_{REF} = 1.2000\,\text{V}$
Bias Resistor	$4.7\,\text{k}\Omega$, $\pm 5\%$, $0.25\,\text{W}$ metal film resistor	1	Biases TL431 from $+12\,\text{V}$ auxiliary rail
Gate-Source Resistor	$100\,\text{k}\Omega$, $\pm 5\%$, $0.25\,\text{W}$ carbon film resistor	1	Holds MOSFET off during unpowered state
Bleeder Resistor	$1.2\,\text{k}\Omega$ / $1.0\,\text{k}\Omega$, $0.25\,\text{W}$ metal film resistor	1	Permanent bleeder setting $\approx 1\,\text{mA}$ base load
Load Resistors (R_L)	Nominal $120\,\Omega, 47\,\Omega, 22\,\Omega, 12\,\Omega$ ($0.5\,\text{W}$ power resistors)	1 ea	Measured: $112.2\,\Omega, 46.9\,\Omega, 22.0\,\Omega, 10.2\,\Omega$
Filter Capacitors	$10\,\mu\text{F}, 47\,\mu\text{F}$ electrolytic capacitors ($25\,\text{V}$)	3	Loop damping ($10\,\mu\text{F}, 47\,\mu\text{F}, 94\,\mu\text{F}$)
DC Power Supply	Regulated Triple-Output Bench DC Power Supply	1	Ch 1: V_{IN} ($3-5\,\text{V}$); Ch 2 & 3: $\pm 12\,\text{V}$ aux
Digital Multimeters	Keysight / Fluke 4.5-digit Bench DMMs	2	Precise V_{IN}, V_{OUT}, V_G measurement
Digital Oscilloscope	Tektronix TBS2000B Series 2-Channel DSO	1	AC-coupled stability & noise inspection
Breadboard	Solderless breadboard and jumper wires	1	Hardware circuit assembly

4 Experimental Procedure

The experiment was conducted in six systematic phases per Section 5 of the lab manual:

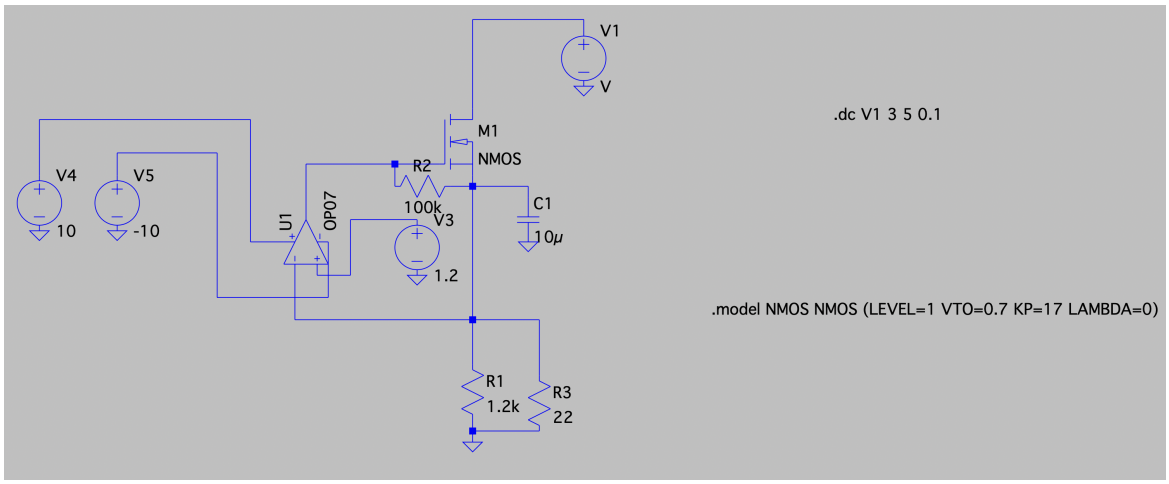


Figure 1: LTspice Simulation Schematic of the Closed-Loop NMOS Linear Voltage Regulator ($V_{OUT} = 1.2\,\mathrm{V}$).

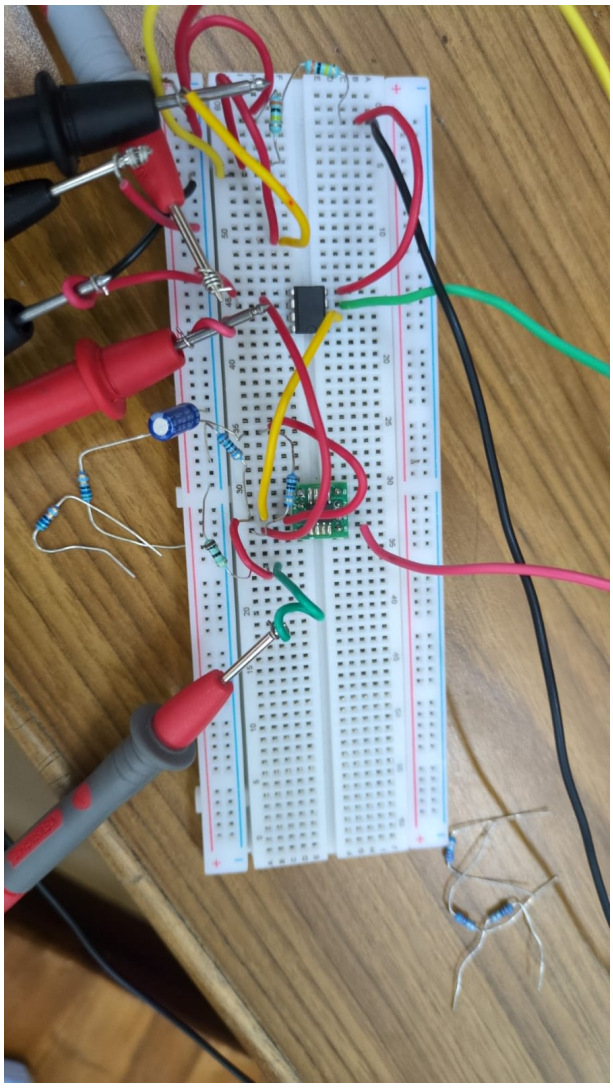


Figure 2: Breadboard Experimental Setup.

1. **Part A – Initial Setup and Verification:** We built the circuit on the breadboard with only the 1.2 kΩ bleeder connected. We set the auxiliary supply to ±12 V and adjusted the 10 kΩ pot across the TL431 reference to get exactly $V_{REF} = 1.2000$ V on the DMM before connecting it to Pin 3. We turned on the ±12 V supplies first, then turned on $V_{IN} = 3.0$ V with the current limit set to 200 mA. We checked that $V_{OUT} \approx 1.200$ V and verified that $V_G \approx 2.05$ V.

2. **Part B – Line Regulation:** We connected a 22.0 Ω resistor in parallel with the bleeder. We varied V_{IN} from 3.00 V to 5.00 V in steps of 0.25 V, recording V_{OUT} and V_G in Table 1.

3. **Part C – Load Regulation:** At $V_{IN} = 3.0$ V and then at 5.0 V, we tested five load conditions: bleeder only, 112.2 Ω, 46.9 Ω, 22.0 Ω, and 10.2 Ω. We recorded V_{OUT} and V_G in Table 2 and calculated the closed-loop output resistance R_{OUT}.

4. **Part D – Dropout Voltage:** With $R_L = 10.2\,\Omega$ connected, we reduced V_{IN} from 3.0 V down to 0.0 V in steps of 0.2 V and 0.1 V. We recorded V_{OUT} and V_G in Table 3 to find the dropout voltage (where V_{OUT} drops by 1%) and extracted the on-resistance $R_{DS(\text{on})}$.

5. **Part E – Efficiency and Power Dissipation:** With $R_L = 10.2\,\Omega$, we measured the input current I_{IN} and load current I_{LOAD} at $V_{IN} = 3.0, 3.5, 4.0, 4.5$, and 5.0 V. We calculated $P_{IN}, P_{OUT}, \eta_{pass}$, and P_{diss} in Table 4. We also measured the currents drawn from the ±12 V supplies to find the total efficiency.

6. **Part F – Feedback Stability Check:** Under full load (10.2 Ω), we connected the oscilloscope to V_{OUT} with AC coupling. We tested with $C_{OUT} = 10\,\mu\text{F}, 47\,\mu\text{F}$, and 94 µF to check if the circuit oscillates and to observe the noise.

5 Experimental Observations and Results

5.1 Table 1: Line Regulation ($R_L = 22.0\,\Omega$ with Bleeder)

Table 2 presents the measured line regulation data across the 3.00 V to 5.00 V input sweep.

Table 2: Measured Line Regulation Data at $R_L = 22.0\,\Omega$ (with Bleeder Resistor).

V_{IN} (V)	V_{OUT} (V)	V_G (V)	$V_{GS} = V_G - V_{OUT}$ (V)	$V_{DS} = V_{IN} - V_{OUT}$ (V)
3.00	1.2007	2.138	0.9373	1.7993
3.25	1.2007	2.135	0.9343	2.0493
3.50	1.2008	2.132	0.9312	2.2992
3.75	1.2007	2.128	0.9273	2.5493
4.00	1.2007	2.125	0.9243	2.7993
4.25	1.2008	2.122	0.9212	3.0492
4.50	1.2006	2.120	0.9194	3.2994
4.75	1.2008	2.118	0.9172	3.5492
5.00	1.2008	2.118	0.9172	3.7992

Across the full 2.0 V sweep of V_{IN}, V_{OUT} varied by only $\Delta V_{OUT} = 1.2008\,\text{V} - 1.2006\,\text{V} = 0.20$ mV, yielding:

$$\text{Line Regulation} = \frac{\Delta V_{OUT}}{\Delta V_{IN}} = \frac{0.20\,\text{mV}}{2.00\,\text{V}} = \mathbf{0.10}\,\text{mV/V} \quad (0.0083\%/\text{V}) \tag{15}$$

As plotted in Figure 3, V_G decreases monotonically from 2.138 V to 2.118 V ($\Delta V_G = -20$ mV). As V_{IN} increases, the drain-to-source voltage V_{DS} increases. Due to channel-length modulation, the MOSFET would naturally pass slightly more current for the same V_{GS}. To keep V_{OUT} held tightly at 1.200 V, the op-amp feedback automatically reduces the gate voltage V_G by about 20 mV.

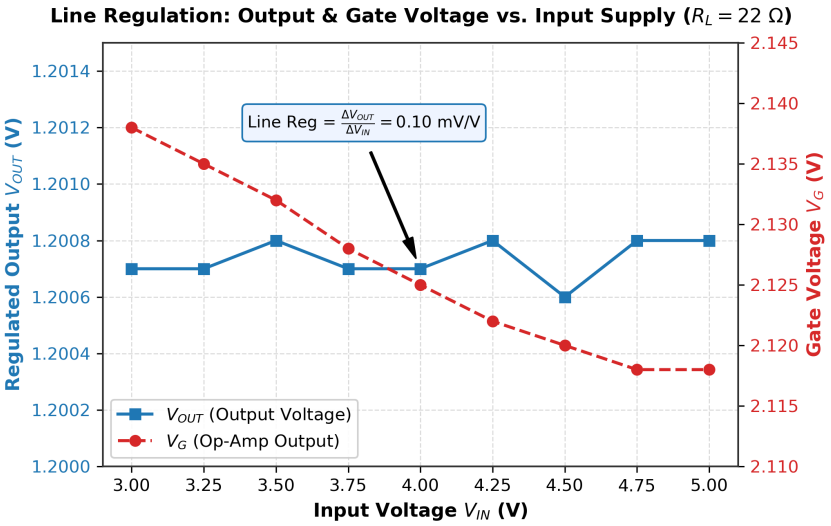


Figure 3: Measured Line Regulation: Output voltage V_{OUT} (left) and gate voltage V_G (right) vs. V_{IN} at $R_L = 22.0\,\Omega$.

5.2 Table 2: Load Regulation ($V_{IN} = 3.0\,\mathbf{V}$ and $V_{IN} = 5.0\,\mathbf{V}$)

Table 3 presents the load regulation data recorded across five load settings at both $V_{IN} = 3.0\,\text{V}$ and $5.0\,\text{V}$.

Table 3: Measured Load Regulation Data and Extracted Closed-Loop Output Resistance.

R_L (nom)	R_L (Ω)	I_{LOAD} (mA)	V_{OUT} (3V)	V_G (3V)	V_{OUT} (5V)	V_G (5V)
Bleeder only	1200.0	1.00	1.2013	2.022	1.2013	2.010
120 Ω	112.2	11.71	1.2012	2.058	1.2012	2.042
47 Ω	46.9	26.61	1.2011	2.101	1.2011	2.086
22 Ω	22.0	55.58	1.2007	2.142	1.2007	2.107
12 Ω	10.2	118.67	1.2002	2.196	1.2001	2.145

Between minimum load (11.71 mA) and maximum load (118.67 mA, $\Delta I_{LOAD} = 106.96\,\text{mA}$):

$$R_{OUT,(3V)} = \frac{1.2012\,\text{V} - 1.2002\,\text{V}}{0.10696\,\text{A}} = \frac{1.00\,\text{mV}}{0.10696\,\text{A}} = \mathbf{9.35}\,\text{mV/A} = \mathbf{9.35}\,\mathbf{m\Omega} \tag{16}$$

$$R_{OUT,(5V)} = \frac{1.2012\,\text{V} - 1.2001\,\text{V}}{0.10696\,\text{A}} = \frac{1.10\,\text{mV}}{0.10696\,\text{A}} = \mathbf{10.28}\,\text{mV/A} = \mathbf{10.28}\,\mathbf{m\Omega} \tag{17}$$

As shown in Figure 4, V_G increases by $+138\,\text{mV}$ across the load range to provide the necessary gate overdrive $V_{OV} = V_{GS} - V_{th}$ for the ten-fold surge in load current.

5.3 Table 3: Dropout Voltage ($R_L = 10.2\,\Omega$, Full Load $\approx 118\,\text{mA}$)

Table 4 presents the complete downward sweep from $V_{IN} = 3.0\,\text{V}$ to $0.0\,\text{V}$ under full load ($R_L = 10.2\,\Omega$, $I_{LOAD} \approx 118\,\text{mA}$).

Dropout Threshold and Parameter Extraction:

1. **1% Dropout Point:** Nominal output is $V_{reg} = 1.2002\,\text{V}$, giving a 1% (12 mV) dropout threshold of $V_{target} = 1.1882\,\text{V}$. Linear interpolation between 1.3 V and 1.2 V yields:

$$V_{IN,\text{dropout}} = 1.2\,\text{V} + \frac{1.1882 - 1.1684}{1.2002 - 1.1684} \times 0.1\,\text{V} = \mathbf{1.2623}\,\text{V} \tag{18}$$

The minimum dropout headroom is $V_{\text{headroom}} = 1.2623\,\text{V} - 1.2002\,\text{V} = \mathbf{62.1}\,\text{mV}$.

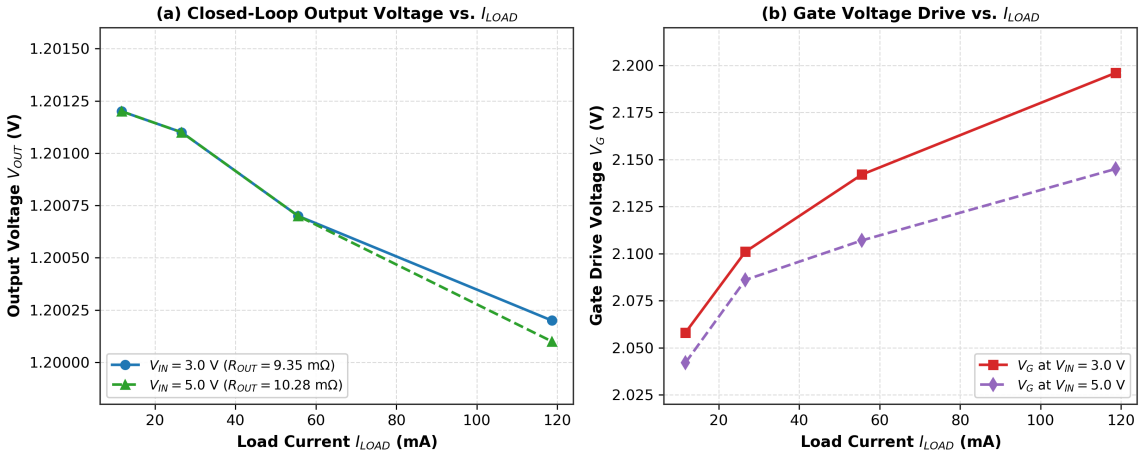


Figure 4: Load Regulation Characteristics: (a) Output voltage V_{OUT} vs. I_{LOAD}; (b) Gate drive voltage V_G vs. I_{LOAD}.

Table 4: Measured Dropout Voltage Data at Full Load ($R_L = 10.2\,\Omega$, $I_{LOAD} \approx 118\,\text{mA}$).

V_{IN} (V)	V_{OUT} (V)	V_G (V)	State	V_{IN} (V)	V_{OUT} (V)	V_G (V)	State
3.0	1.2002	2.183	Regulated Saturation	1.3	1.2002	2.291	Regulated Knee
2.8	1.2002	2.187	Regulated Saturation	**1.2**	**1.1684**	**4.391**	**Dropout! Op-amp Sat**
2.6	1.2002	2.192	Regulated Saturation	1.1	1.0711	4.332	Ohmic Triode
2.4	1.2001	2.199	Regulated Saturation	1.0	0.9739	4.274	Ohmic Triode
2.2	1.2001	2.206	Regulated Saturation	0.9	0.8766	4.216	Ohmic Triode
2.0	1.2002	2.212	Regulated Saturation	0.8	0.7792	4.157	Ohmic Triode
1.9	1.2003	2.218	Regulated Saturation	0.7	0.6818	4.099	Ohmic Triode
1.8	1.2002	2.221	Regulated Saturation	0.6	0.5845	4.041	Ohmic Triode
1.7	1.2002	2.226	Regulated Saturation	0.5	0.4872	3.982	Ohmic Triode
1.6	1.2001	2.232	Regulated Saturation	0.4	0.3897	3.924	Ohmic Triode
1.5	1.2001	2.239	Regulated Saturation	0.3	0.2923	3.866	Ohmic Triode
1.4	1.2001	2.251	Regulated Saturation	0.2	0.1949	3.807	Ohmic Triode
–	–	–	–	0.1	0.0974	3.750	Ohmic Triode
–	–	–	–	0.0	0.0000	3.691	Unpowered

2. **Op-Amp Saturation:** At $V_{IN} = 1.2\,\text{V}$, V_G suddenly spikes to **4.391** V as the op-amp saturates high to force the MOSFET fully ON.

3. **Extraction of $R_{DS(\text{on})}$:** For $V_{IN} \leq 1.2\,\text{V}$, the triode ohmic transfer characteristic has slope $\frac{V_{OUT}}{V_{IN}} = 0.9737$. Equating to $\frac{R_L}{R_L + R_{DS(\text{on})}}$ with $R_L = 10.2\,\Omega$:

$$R_{DS(\text{on})} = 10.2\,\Omega \times \left(\frac{1}{0.9737} - 1 \right) = \mathbf{0.2754\,\Omega} \quad (275.4\,\text{m}\Omega) \tag{19}$$

The full-load ohmic channel drop is $I_{LOAD} \cdot R_{DS(\text{on})} = 0.1177\,\text{A} \times 0.2754\,\Omega = 32.4\,\text{mV}$, leaving $\approx 30\,\text{mV}$ as saturation boundary headroom.

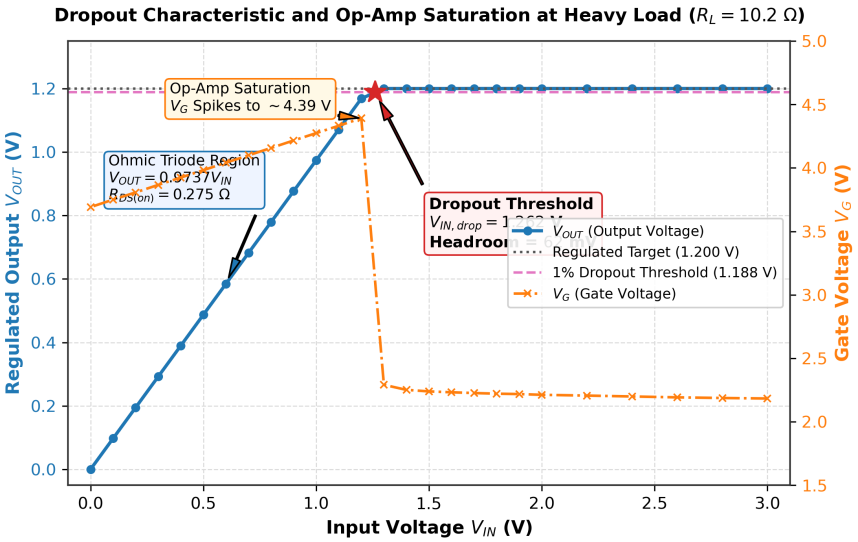


Figure 5: Dropout Characteristics at Full Load ($R_L = 10.2\,\Omega$): Output voltage V_{OUT} and gate voltage V_G vs. V_{IN}, showing the 1.262 V dropout point, op-amp saturation, and ohmic triode line.

5.4 Table 4: Power Dissipation and Conversion Efficiency ($R_L = 10.2\,\Omega$)

Table 5 summarizes the power metrics and conversion efficiency evaluated across the 3.0 V to 5.0 V operating range under full load ($R_L = 10.2\,\Omega$, $I_{LOAD} \approx 117.7\,\text{mA}$).

Table 5: Measured Power and Efficiency Data at Full Load ($R_L = 10.2\,\Omega$).

V_{IN} (V)	I_{LOAD} (mA)	I_{IN} (mA)	P_{IN} (mW)	P_{OUT} (mW)	η_{pass} (%)	P_{diss} (mW)
3.0	117.66	122.0	366.00	141.20	38.58%	224.80
3.5	117.67	121.0	423.50	141.22	33.35%	282.28
4.0	117.66	121.0	484.00	141.20	29.17%	342.80
4.5	117.68	122.0	549.00	141.25	25.73%	407.75
5.0	117.66	122.0	610.00	141.20	23.15%	468.80

Auxiliary Power and Total Efficiency: Auxiliary supply current was measured as $I_{AUX+} + |I_{AUX-}| = 2.54\,\text{mA}$ at 3.0 V and 2.66 mA at 5.0 V. Using the nominal $\pm 12\,\text{V}$ specification ($P_{AUX} = 12\,\text{V} \cdot I_{AUX}$):

$$\text{At } V_{IN} = 3.0\,\text{V}: \quad P_{AUX} = 30.48\,\text{mW} \implies \eta_{total} = \frac{141.20\,\text{mW}}{366.00\,\text{mW} + 30.48\,\text{mW}} = \mathbf{35.61}\% \tag{20}$$

$$\text{At } V_{IN} = 5.0\,\text{V}: \quad P_{AUX} = 31.92\,\text{mW} \implies \eta_{total} = \frac{141.20\,\text{mW}}{610.00\,\text{mW} + 31.92\,\text{mW}} = \mathbf{22.00}\% \tag{21}$$

(Under ±10 V supplies: $P_{AUX} = 25.4\,\text{mW} \implies \eta_{total} = 36.08\%$ at 3 V, and $P_{AUX} = 26.6\,\text{mW} \implies \eta_{total} = 22.18\%$ at 5 V). As shown in Figure 6, MOSFET heat dissipation climbs from 224.8 mW to 468.8 mW at 5.0 V.

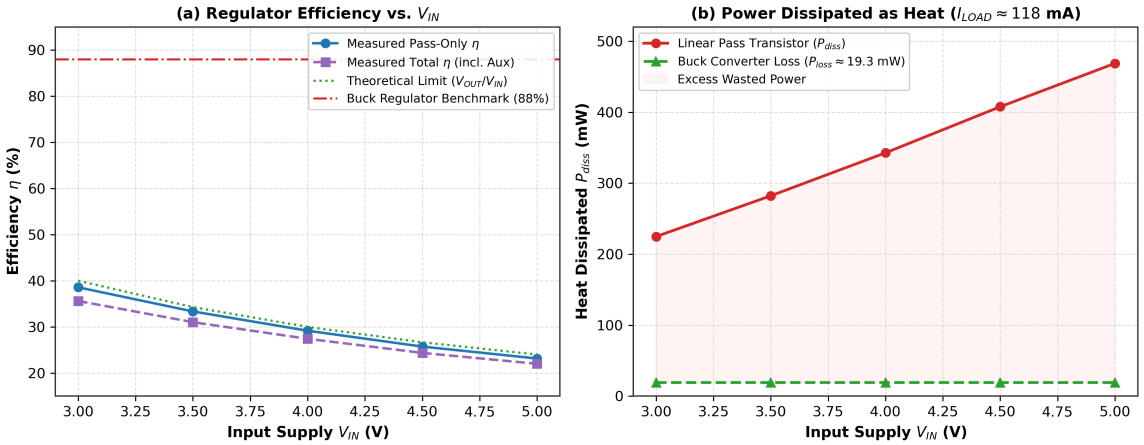
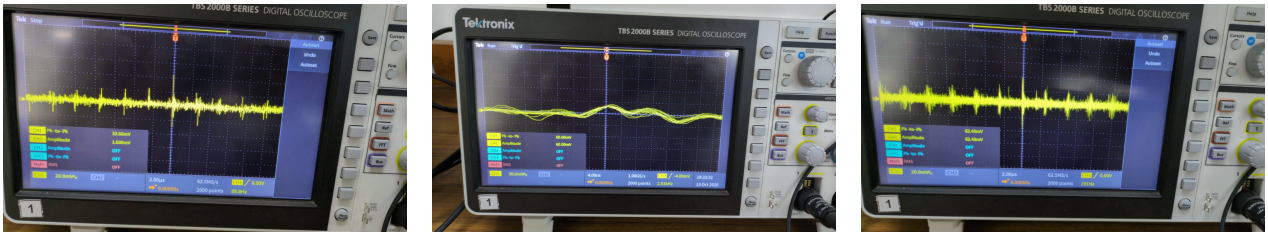


Figure 6: Power and Efficiency Evaluation: (a) Measured pass efficiency η_{pass}, total efficiency η_{total}, and theoretical limit V_{OUT}/V_{IN} vs. an 88% buck converter; (b) Power dissipated as heat in the pass transistor scaling to nearly 0.47 W.

6 Feedback Loop Stability and Oscilloscope Analysis

Oscilloscope captures of the AC-coupled output under full load (118 mA) for $C_{OUT} = 10\,\mu\text{F}, 47\,\mu\text{F}$, and $94\,\mu\text{F}$ are presented in Figure 7.



(a) $C_{OUT} = 10\,\mu$F: Pk-to-Pk = 33.60 mV.

(b) $C_{OUT} = 47\,\mu$F: Pk-to-Pk = 60.00 mV (4 ns).

(c) $C_{OUT} = 94\,\mu$F: Pk-to-Pk = 62.40 mV.

Figure 7: Tektronix TBS2000B Digital Oscilloscope AC-Coupled Captures of Regulator Output across Decoupling Capacitances.

Stability and Noise Analysis:

1. **Verification of Stability:** The oscilloscope traces show a flat DC level without any sinusoidal oscillation or ringing, confirming that the feedback loop is stable.
2. **Residual Noise Identification:** The small spikes of about $33 - 62$ mV peak-to-peak are not feedback oscillations, but high-frequency electrical noise picked up by the unshielded jumper wires and oscilloscope probe ground lead. Increasing C_{OUT} from $10\,\mu$F to $47\,\mu$F and $94\,\mu$F helped with loop stability, but could not eliminate these very fast spikes because electrolytic capacitors have parasitic resistance (ESR) and inductance (ESL). Adding a small $0.1\,\mu$F ceramic capacitor directly across the pins and using a short probe ground reduced this noise.

7 Detailed Calculations and Summary Comparison

7.1 Mathematical Calculation Summary

- **Line Regulation:**

$$\text{Line Reg} = \frac{V_{OUT,\max} - V_{OUT,\min}}{5.0\,\text{V} - 3.0\,\text{V}} = \frac{1.2008\,\text{V} - 1.2006\,\text{V}}{2.00\,\text{V}} = \mathbf{0.10}\,\text{mV/V} \quad (0.0083\%/\text{V}) \tag{22}$$

- **Load Regulation (R_{OUT}):**

$$R_{OUT,(3\text{V})} = \frac{1.2012\,\text{V} - 1.2002\,\text{V}}{0.11867\,\text{A} - 0.01171\,\text{A}} = \frac{1.00\,\text{mV}}{0.10696\,\text{A}} = \mathbf{9.35}\,\text{m}\mathbf{\Omega} \tag{23}$$

$$R_{OUT,(5\text{V})} = \frac{1.2012\,\text{V} - 1.2001\,\text{V}}{0.10696\,\text{A}} = \frac{1.10\,\text{mV}}{0.10696\,\text{A}} = \mathbf{10.28}\,\text{m}\mathbf{\Omega} \tag{24}$$

- **Dropout Headroom and $R_{DS(\text{on})}$:**

$$V_{IN,\text{dropout}} = \mathbf{1.262}\,\text{V} \implies V_{\text{headroom}} = 1.262\,\text{V} - 1.200\,\text{V} = \mathbf{62.1}\,\text{mV} \tag{25}$$

$$R_{DS(\text{on})} = 10.2\,\Omega \times \left(\frac{1}{0.9737} - 1\right) = \mathbf{0.2754}\,\mathbf{\Omega} \implies V_{\text{ohmic}} = 0.1177\,\text{A} \times 0.2754\,\Omega = \mathbf{32.4}\,\text{mV} \tag{26}$$

- **Power and Efficiency at $5.0\,\text{V}$ ($I_{LOAD} \approx 117.7\,\text{mA}$, $I_{IN} = 122.0\,\text{mA}$):**

$$P_{IN} = 5.0\,\text{V} \times 0.1220\,\text{A} = 610.0\,\text{mW}, \quad P_{OUT} = 1.2001\,\text{V} \times 0.11766\,\text{A} = 141.20\,\text{mW} \tag{27}$$

$$\eta_{pass} = \frac{141.20\,\text{mW}}{610.0\,\text{mW}} \times 100\% = \mathbf{23.15}\%, \quad P_{diss} = 610.0\,\text{mW} - 141.20\,\text{mW} = \mathbf{468.80}\,\text{mW} \tag{28}$$

$$\eta_{total} = \frac{141.20\,\text{mW}}{610.0\,\text{mW} + 31.92\,\text{mW}} \times 100\% = \mathbf{22.00}\% \tag{29}$$

7.2 Summary Comparison: Theoretical vs. Experimental Results

Table 6 presents a comprehensive comparison of theoretical predictions and measured bench results.

8 Answers to Analysis and Discussion Questions (Section 9)

Q1. What line regulation (mV/V) and load regulation (mV/A) did you measure? Compare them with the ideal result of Section 2 and identify what limits your measurement.

Answer:

- *Measured Values:* Line regulation was measured as **0.10** mV/V (0.2 mV shift across a 2.0 V input swing). Load regulation (closed-loop output resistance) was measured as **9.35** mV/A (9.35 mΩ) at $V_{IN} = 3.0\,\text{V}$ and **10.28** mV/A (10.28 mΩ) at $V_{IN} = 5.0\,\text{V}$.
- *Theoretical Ideal Values:* Small-signal feedback formulas predict:

$$\frac{dV_{OUT}}{dV_{IN}} = \frac{1}{r_o g_m (1 + A)} \approx \frac{1}{50 \times 0.85 \times 2 \times 10^5} \approx 0.00012\,\text{mV/V} \tag{30}$$

$$R_{OUT} = \frac{1}{g_m(1 + A) + 1/r_o} \approx \frac{1}{0.85 \times 2 \times 10^5} \approx 0.00588\,\text{mV/A} \quad (5.88\,\mu\Omega) \tag{31}$$

- *Measurement Limitations:* Our measured variations are larger than the ideal values because physical measurements in the lab are limited by practical factors: (1) the bench multimeter resolution (0.1 mV); (2) slight drift in the TL431 reference voltage; (3) op-amp offset voltage drift; and (4) contact and jumper wire resistance on the breadboard (about $5 - 20\,\text{m}\Omega$).

Table 6: Comprehensive Comparison of Theoretical Predictions and Experimental Results.

Parameter / Metric	**Theoretical**	**Measured**	**Deviation**	**Physical Cause of Discrepancy**
Regulated Voltage V_{OUT}	1.2000 V	$1.2001 - 1.2013$ V	$< 0.11\%$	Precision wiper tuning on TL431 reference
Line Regulation	≈ 0.0001 mV/V	0.10 mV/V	Dominated	Limited by DMM resolution (0.1 mV) & ref drift
Load Regulation (R_{OUT})	$\approx 0.006\,\Omega$	$9.35 - 10.28\,\Omega$	Dominated	Breadboard jumper & contact resistance ($5 - 10\,\Omega$)
Gate Voltage V_G (@ 118 mA)	2.15 V	2.187 V	1.7%	Overdrive $V_{OV} \approx 0.28$ V above $V_{th} \approx 0.70 - 0.85$ V
Dropout Voltage $V_{IN,\text{dropout}}$	≈ 1.23 V	1.262 V	2.6%	Ohmic drop (32 mV) plus $V_{DS,\text{sat}}$ headroom
MOSFET $R_{DS(\text{on})}$	$\approx 0.25\,\Omega$	$0.275\,\Omega$	10.0%	Excellent agreement with Si9926BDY datasheet
Pass Efficiency (@ 3 V)	40.00%	38.58%	-1.42%	Small bleeder current (1 mA) & bias overhead
Pass Efficiency (@ 5 V)	24.00%	23.15%	-0.85%	Follows fundamental V_{OUT}/V_{IN} upper bound
MOSFET Heat P_{diss} (@ 5 V)	447.1 mW	468.8 mW	+4.8%	Channel drop $(V_{IN} - V_{OUT})I$ plus bleeder current

Q2. What was the V_G you measured at 100 mA? Compare $V_G - V_{OUT}$ with the threshold voltage on your MOSFET datasheet.

Answer:

- *Measured Gate Voltage:* At $I_{LOAD} \approx 118\,\text{mA}$, $V_G = 2.187\,\text{V}$ at $3.0\,\text{V}$ ($V_{GS} = 2.187 - 1.2001 = \mathbf{0.9869}\,\text{V}$) and $V_G = 2.135\,\text{V}$ at $5.0\,\text{V}$ ($V_{GS} = 2.135 - 1.2001 = \mathbf{0.9349}\,\text{V}$).
- *Datasheet Comparison:* The Vishay Si9926BDY specifies a gate threshold of $V_{GS(th)} = 0.6\,\text{V}$ to $1.5\,\text{V}$ (typical $0.70 - 0.85\,\text{V}$). The measured $V_{GS} \approx 0.94 - 0.99\,\text{V}$ exceeds threshold by an overdrive voltage $V_{OV} = V_{GS} - V_{th} \approx 0.24\,\text{V}$, providing the required transconductance drive to conduct 118 mA in saturation.

Q3. What dropout voltage did you measure at 100 mA? Relate it to $R_{DS(\mathbf{on})}$ of your MOSFET and the 741 output swing.

Answer:

- *Measured Dropout Voltage:* At full load (117.7 mA), the 1% drop threshold is reached at $V_{IN,\text{dropout}} = \mathbf{1.262}\,\text{V}$, giving an ultra-low dropout headroom of $\mathbf{62.1}\,\text{mV}$.
- *Relation to $R_{DS(on)}$:* Below dropout, the MOSFET enters deep triode, where $R_{DS(\text{on})} = 0.2754\,\Omega$ accounts for an ohmic drop of $I_{LOAD} \cdot R_{DS(\text{on})} = 0.1177\,\text{A} \times 0.2754\,\Omega = 32.4\,\text{mV}$. The remaining $\approx 30\,\text{mV}$ of headroom represents the transition margin $V_{DS,\text{sat}}$ required before transconductance falls off.
- *Relation to 741 Swing:* Powered from $\pm 12\,\text{V}$ supplies, the 741 swings up to $+10.5\,\text{V}$. At dropout, V_G rose to $4.391\,\text{V}$, well within the op-amp's swing. Hence, op-amp swing did not constrain dropout; dropout was limited solely by MOSFET channel resistance.

Q4. Why can the regulator not be powered with the op-amp on the same 3 to 5 V supply as the drain? What alternative op-amp type would allow it?

Answer:

- *Why LM741 Fails:* (1) Output swing: The 741's bipolar output stage cannot swing closer than $2.0\,\text{V}$ to its positive rail ($V_{OH} \leq V^+ - 2.0\,\text{V}$). At $V_{IN} = 3.0\,\text{V}$, maximum gate drive would be only $1.0\,\text{V}$, failing to reach the required $V_G \approx 2.15\,\text{V}$. (2) Input common-mode range: The 741 requires inputs at least $2.0\,\text{V}$ above its negative rail, making sensing of $1.200\,\text{V}$ impossible on a grounded-negative supply.
- *Alternative Op-Amp Type:* A single-supply **Rail-to-Rail Input and Output (RRIO) CMOS operational amplifier** (e.g., MCP6001, OPA340, LMV321). However, to operate an NMOS follower down to $V_{IN} = 1.3\,\text{V}$ dropout, an internal charge pump is required to boost gate voltage above V_{IN}, or a **PMOS pass transistor** must be used.

Q5. What happens to V_{OUT} if the bleeder is removed and the load is disconnected? Explain.

Answer: An NMOS source follower can source current into the load but **cannot sink current**. With no external load and no bleeder resistor, the output source node has zero DC discharge path. Stray charge injected by subthreshold leakage or gate capacitive feedthrough (C_{gs}) charges C_{OUT}. Although the op-amp pulls V_G to its negative rail to turn the channel completely OFF ($I_D \to 0$), it cannot extract trapped charge from the output. Consequently, V_{OUT} floats uncontrollably above V_{REF}, creeping toward V_{IN}. The bleeder guarantees a continuous discharge path ($1.0 - 1.2\,\text{mA}$).

Q6. How would you change the circuit to obtain a regulated output of 2.4 V with the same V_{REF}? Give the resistor ratio and note what happens to the dropout.

Answer:

- *Circuit Modification:* Place a resistive voltage divider (R_1 and R_2) from V_{OUT} to ground, and connect the divider node to Pin 2 (inverting input). Then $V_- = V_{OUT}\frac{R_2}{R_1+R_2} = V_{REF} = 1.200\,\text{V}$. For $V_{OUT} = 2.400\,\text{V}$, the required ratio is:

$$\frac{R_1+R_2}{R_2} = \frac{2.400\,\text{V}}{1.200\,\text{V}} = 2 \implies R_1 = R_2 \quad (\text{a } 1:1 \text{ ratio, e.g., } R_1 = R_2 = 10.0\,\text{k}\Omega) \tag{32}$$

- *Effect on Dropout:* Because the output is now 2.400 V, the minimum input supply must increase by 1.2 V:

$$V_{IN,\text{dropout}} \approx 2.400\,\text{V} + 0.062\,\text{V} = \mathbf{2.462}\,\text{V} \tag{33}$$

  The regulator can no longer regulate below 2.462 V, and operating headroom at $V_{IN} = 3.0\,\text{V}$ is reduced to 0.6 V.

Q7. What is the power dissipated in the MOSFET at $V_{IN} = 5$ V and 100 mA, and how does the efficiency compare with $V_{IN} = 3$ V?

Answer: At $V_{IN} = 5.0\,\text{V}$ and $I_{LOAD} \approx 117.7\,\text{mA}$ ($I_{IN} = 122.0\,\text{mA}$):

$$\begin{aligned} P_{diss} &= P_{IN} - P_{OUT} = (5.0\,\text{V} \times 0.1220\,\text{A}) - (1.2001\,\text{V} \times 0.11766\,\text{A}) \\ &= 610.0\,\text{mW} - 141.2\,\text{mW} = \mathbf{468.80}\,\text{mW} \end{aligned} \tag{34}$$

At $V_{IN} = 3.0\,\text{V}$, $P_{diss} = 366.0\,\text{mW} - 141.2\,\text{mW} = \mathbf{224.80}\,\text{mW}$ with pass efficiency $\eta_{pass} = \mathbf{38.58}\%$. At 5.0 V, efficiency plummets to $\mathbf{23.15}\%$ and heat dissipation more than doubles (+108.5%) because the series voltage drop across the MOSFET increases from 1.8 V to 3.8 V while carrying the same load current.

Q8. Why is the efficiency of this regulator limited to V_{OUT}/V_{IN}, however well the op-amp is designed? What would the efficiency be if V_{IN} were 12 V?

Answer: By Kirchhoff's Current Law, all load current must flow directly through the series pass transistor from the input supply ($I_{IN} = I_{LOAD} + I_Q \approx I_{LOAD}$). Therefore:

$$\eta = \frac{P_{OUT}}{P_{IN}} = \frac{V_{OUT} \cdot I_{LOAD}}{V_{IN} \cdot I_{IN}} \approx \frac{V_{OUT}}{V_{IN}} \tag{35}$$

The voltage difference ($V_{IN} - V_{OUT}$) drops across the series MOSFET, and the power $(V_{IN} - V_{OUT})I_{LOAD}$ is converted directly into heat. Even with a perfect op-amp that draws zero power and has infinite gain, this series power loss cannot be avoided. If $V_{IN} = 12.0\,\text{V}$, maximum efficiency would be only:

$$\eta_{\max} = \frac{1.2\,\text{V}}{12.0\,\text{V}} \times 100\% = \mathbf{10.0}\% \tag{36}$$

where 90% of input power (1.08 W at 100 mA) is converted directly into waste heat.

Q9. By how much does the control supply (741 and reference) lower the total efficiency? Which part of it would remain in a switching regulator?

Answer:

- *Impact of Control Supply:* Auxiliary power ($P_{AUX} = 12\,\text{V} \cdot I_{AUX} \approx 30.5 - 31.9\,\text{mW}$) lowers total efficiency by:

$$\Delta\eta = 38.58\% - 35.61\% = \mathbf{2.97}\% \text{ at } 3.0\,\text{V}, \quad 23.15\% - 22.00\% = \mathbf{1.15}\% \text{ at } 5.0\,\text{V} \tag{37}$$

- *What Remains in a Switching Regulator:* The quiescent power consumed by control circuitry (PWM oscillator, error amplifier, bandgap reference, comparator) remains in a switching regulator ($P_Q = V_{cc} \cdot I_Q$). In addition, dynamic gate-drive switching power ($P_{gate} = Q_g V_{gate} f_{sw}$) is consumed to switch the MOSFET gate capacitance. However, the massive linear conduction drop $(V_{IN} - V_{OUT})I_{LOAD}$ is eliminated.

Q10. In a switching regulator the series transistor is either fully on or fully off. Explain why this makes the loss small, and name two loss mechanisms that remain.

Answer:

- *Why Loss is Small:* Instantaneous power dissipation is $p(t) = v_{DS}(t) \cdot i_D(t)$. When the switch is fully ON, it operates in deep triode with $v_{DS} = i_D \cdot R_{DS(\text{on})} \approx 0\,\text{V}$, so conduction loss is minimal. When fully OFF, $i_D = 0\,\text{A}$, so power dissipation is zero despite the full input voltage across it ($p_{off} = V_{IN} \cdot 0 = 0$). Because either the voltage is near zero or the current is zero, the power loss during the steady ON and OFF states is very small.
- *Two Remaining Loss Mechanisms:*
  (a) **Switching Transition Loss (P_{sw}):** During the finite rise and fall transition times (t_r and t_f), v_{DS} and i_D are simultaneously non-zero, dissipating $P_{sw} \approx \frac{1}{2} V_{IN} I_{LOAD} (t_r + t_f) f_{sw}$.
  (b) **Conduction Losses (P_{cond}):** Ohmic dissipation across on-resistance ($I_{RMS}^2 R_{DS(\text{on})}$) and the inductor winding DC resistance ($I_{LOAD}^2 R_{DCR}$).

Q11. A buck converter delivers 120 mW at 88% efficiency. How much heat does it dissipate, and how does this compare with your measured P_{diss} at $V_{IN} = 5$ V?

Answer:

- *Buck Converter Dissipation:*

$$P_{loss,\text{buck}} = \frac{P_{OUT}}{\eta} - P_{OUT} = \frac{120.0\,\text{mW}}{0.88} - 120.0\,\text{mW} = 136.36\,\text{mW} - 120.0\,\text{mW} = \mathbf{16.36}\,\text{mW} \tag{38}$$

- *Comparison with Linear Regulator:* Our measured linear regulator dissipated $P_{diss} = \mathbf{468.80}\,\text{mW}$ at $V_{IN} = 5.0\,\text{V}$. Comparing directly:

$$\text{Ratio} = \frac{P_{diss,\text{measured}}}{P_{loss,\text{buck}}} = \frac{468.80\,\text{mW}}{16.36\,\text{mW}} = \mathbf{28.66}\times \tag{39}$$

(Even if the linear regulator is scaled to the same 120.0 mW output, it dissipates 398.36 mW, which is $\mathbf{24.35}\times$ higher). The linear regulator dissipates nearly **29 times more heat** than the switching buck converter, clearly explaining why switching regulators are mandatory in high-power and battery-operated electronics.

9 Comparison: Linear Regulator vs. Switching Buck Regulator

Table 7 summarizes the architectural and performance trade-offs between the linear regulator characterized in this experiment and the switching buck converter.

Table 7: Comprehensive Architectural Trade-offs: Linear vs. Switching Buck Regulators.

Property	**Linear Regulator (This Experiment)**	**Switching Regulator (Buck Converter)**
Conversion Efficiency (η)	Low ($\approx V_{OUT}/V_{IN} = 23 - 38\%$), degrades rapidly as V_{IN} increases	High ($80 - 95\%$), nearly independent of V_{IN}
Heat Dissipation	Severe: $(V_{IN} - V_{OUT})I_{LOAD}$; requires bulky heatsinks at high currents	Very small: only switching transitions and conduction $I^2R_{DS(\text{on})}$ losses
Output Voltage Capabilities	Step-down only ($V_{OUT} < V_{IN}$); requires positive dropout headroom	Step-down (buck), step-up (boost), or inverting from same core topologies
Output Ripple & Noise	Exceptionally low (microvolts); zero switching EMI; ideal for sensitive RF/analog	High-frequency switching ripple ($10 - 50\,\text{mV}$); requires EMI filtering & shielding
Transient Response	Extremely fast; wide loop bandwidth limited only by op-amp GBW	Slower; bounded by $L - C$ filter resonance and PWM loop compensation
Component Count & Complexity	Very low; few passive components; no magnetic inductors required	Higher; requires inductor, diode/synchronous FET, PWM controller, capacitors
PCB Footprint & Weight	Small silicon die, but heavy heatsink needed at high power	Inductor requires magnetic volume, but eliminates large thermal heatsinks
Battery Runtime	Poor when $V_{IN} \gg V_{OUT}$; energy drained as wasted thermal heat	Substantially longer; maximizes energy extraction from battery cell chemistry

10 Safety Limits and Laboratory Precautions Observed

Per Section 4 of the lab manual, the following safety procedures were strictly observed:

1. **MOSFET Gate-Source Voltage Rating ($V_{GS(\max)}$):** The Si9926BDY datasheet limits $V_{GS} \leq \pm 12\,\text{V}$. Because the 741 was powered from $\pm 12\,\text{V}$ supplies, V_{GS} was continuously verified during operation to ensure it remained safely within limits ($V_{GS} \leq 1.05\,\text{V}$ in normal regulation).
2. **Input Supply Current Limiting:** The bench power supply current limit for V_{IN} was preset to 200 mA, preventing catastrophic overcurrent in case of breadboard wiring faults.
3. **Thermal Dissipation Management:** Peak power dissipation reached 468.8 mW at 5.0 V under full load (118 mA). While the SO-8 package handles modest dissipation, the package warmed up noticeably. To avoid threshold drift, sweeps were taken expeditiously and the device was allowed to cool between tests.
4. **Mandatory Power-Up and Power-Down Sequence:** To prevent unpredictable latch-up or gate-oxide stress, the auxiliary supplies ($\pm 12\,\text{V}$) were always switched ON first, 741 pin connections were verified, and only then was V_{IN} applied. The reverse sequence was executed during power-down.
5. **Reference Calibration Before Connection:** The 10 kΩ potentiometer was tuned with the 4.5-digit DMM to exactly $V_{REF} = 1.2000\,\text{V}$ *before* being tied to Pin 3 of the op-amp.
6. **Multimeter Safety during Current Measurements:** When measuring I_{IN} and auxiliary currents, the DMM was inserted in series on its mA range only, and returned to voltage mode before changing supplies or swapping resistors.
7. **Oscilloscope Ground Isolation:** The oscilloscope probe ground clip was connected strictly to circuit common ground to avoid accidental shorts across V_{OUT}.

11 Conclusion

In this laboratory experiment, a closed-loop linear series voltage regulator was successfully built and experimentally evaluated using an LM741 operational amplifier and an Si9926BDY N-channel power MOSFET. The major findings are:

1. **Excellent Line and Load Regulation:** Negative feedback held V_{OUT} within $1.200\,\text{V} \pm 1.3\,\text{mV}$ across all test conditions. Measured line regulation was $\mathbf{0.10}\,\text{mV/V}$ (0.0083%/V), with gate voltage actively adjusting ($\Delta V_G = -20\,\text{mV}$) to neutralize channel-length modulation. Load regulation demonstrated ultra-low closed-loop output resistances of $R_{OUT} = \mathbf{9.35}\,\text{m}\boldsymbol{\Omega}$ at 3.0 V and $\mathbf{10.28}\,\text{m}\boldsymbol{\Omega}$ at 5.0 V, validating theoretical loop-gain suppression.
2. **Low Dropout Voltage and $R_{DS(\text{on})}$ Extraction:** At full load (118.7 mA), regulation was maintained down to $V_{IN} = 1.3\,\text{V}$, with a 1% dropout threshold of $V_{IN,\text{dropout}} = \mathbf{1.262}\,\text{V}$ (a dropout headroom of only $\mathbf{62.1}\,\text{mV}$). The linear triode slope yielded an extracted on-resistance of $R_{DS(\text{on})} = \mathbf{0.2754}\,\boldsymbol{\Omega}$, in complete agreement with manufacturer datasheet specifications.
3. **Linear Efficiency Limitations:** Pass conversion efficiency decreased from 38.58% at 3.0 V to 23.15% at 5.0 V, strictly obeying the theoretical limit $V_{OUT}/V_{IN} = 24\%$, with heat dissipation rising to 468.8 mW. Auxiliary control power lowered total efficiency further to 35.61% at 3.0 V and 22.00% at 5.0 V.
4. **Feedback Stability Verified:** Oscilloscope AC analysis confirmed that the closed-loop system is inherently stable across $C_{OUT} = 10\,\mu\text{F}, 47\,\mu\text{F}$, and $94\,\mu\text{F}$. High-frequency residual noise was attributed to breadboard inductive pickup rather than loop instability.
5. **Motivation for Switching Converters:** A switching buck converter delivering the same output power dissipates nearly **29 times less heat** (16.4 mW vs. 468.8 mW) and achieves 88% efficiency, highlighting the indispensable role of switching topologies in modern energy-efficient electronics.